

HW 7 (Mastermath HoTT)

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Problem 1. Pullbacks.

- (a) (Pullback pasting lemma.) Consider a diagram of the following form in a category $\mathcal{C}$.

$$\begin{array}{ccccc} \bullet & \longrightarrow & \bullet & \longrightarrow & \bullet \\ \downarrow & & \downarrow & & \downarrow \\ \bullet & \longrightarrow & \bullet & \longrightarrow & \bullet \end{array}$$

Suppose the right square is a pullback. Show that the outer rectangle is a pullback if and only if the left square is a pullback.

- (b) (Sections from diagonals.) Consider a pullback square

$$\begin{array}{ccc} P & \xrightarrow{g} & B \\ \downarrow f \lrcorner & & \downarrow h \\ A & \xrightarrow{k} & C \end{array}$$

in a category $\mathcal{C}$. Show that there is a bijection between sections of f (i.e. morphisms $s : A \rightarrow P$ with $fs = 1_A$) and diagonals of the square (i.e. morphisms $d : A \rightarrow B$ with $hd = k$).

- (c) Let $\mathcal{C}$ be a category with a terminal object $*$ and binary products. Show that for any object A of $\mathcal{C}$, the following square is a pullback.

$$\begin{array}{ccc} A \times A & \xrightarrow{\pi_1} & A \\ \downarrow \pi_0 & & \downarrow !_A \\ A & \xrightarrow{!_A} & * \end{array}$$

- (d) Let $\mathcal{C}$ be a category with binary products. Show that for any morphism $f : X \rightarrow Y$ in $\mathcal{C}$, the following square is a pullback.

$$\begin{array}{ccc} X \times Y & \xrightarrow{f \times 1_Y} & Y \times Y \\ \downarrow \pi_0 & & \downarrow \pi_0 \\ X & \xrightarrow{f} & Y \end{array}$$

Recall the following definition.

Definition 0.1. Given a choice of path object $\text{Path } A$ for each object A , we can produce a *mapping path space* factorization on the category.

We take a morphism $f : X \rightarrow Y$ to the following factorization

$$X \xrightarrow{\lambda_f} X \times_{\bullet} \text{Path } Y \xrightarrow{\epsilon_f} Y$$

where $X \times_{\bullet} \text{Path } Y$ is the following pullback

$$\begin{array}{ccc} X \times_{\bullet} \text{Path } Y & \longrightarrow & \text{Path } Y \\ \downarrow & \lrcorner & \downarrow \pi_0 \epsilon \\ X & \xrightarrow{f} & Y \end{array}$$

The map ϵ_f is given by projecting to $\text{Path } Y$ and then taking $\pi_1 \epsilon$. The map λ_f is induced by 1_X and $r_Y f$.

Problem 2. (The right component of the mapping path space factorization is a display map.) Suppose that $\mathcal{C}$ is a clan (Def. 2.7 in the notes) with a choice of path objects for each object, satisfying that each $\epsilon : \text{Path } Y \rightarrow Y \times Y$ is in $\mathcal{D}$. Show that ϵ_f is in $\mathcal{D}$. (Hint: Use Problem 1 (a), (c) and (d))

Problem 3. (wfs induces a clan on the fibrant objects.) Consider a weak factorization system $(\mathcal{L}, \mathcal{R})$ on a category $\mathcal{C}$ with finite limits. Let $\mathcal{C}_{\mathcal{F}}$ denote the full subcategory of $\mathcal{C}$ spanned by the objects X such that the unique map $X \rightarrow *$ is in $\mathcal{R}$. Show that:

- (i) The wfs $(\mathcal{L}, \mathcal{R})$ induces a wfs $(\mathcal{L}_{\mathcal{F}}, \mathcal{R}_{\mathcal{F}})$ on $\mathcal{C}_{\mathcal{F}}$.
- (ii) $(\mathcal{C}_{\mathcal{F}}, \mathcal{R}_{\mathcal{F}})$ is a clan.